



Windows Vista basically accomplishes what Linux has been doing for a long time—ensure that regular users don't have access to critical operating system files and services

All this is very leading, but it really doesn't say *anything*. IIS servers are often taken down by exploiting vulnerabilities in *Windows* (an unfortunate side-effect of Windows' integration with its software; more on this later), and not necessarily in IIS itself. Secondly, while the Netcraft list of sites with the longest uptime is dominated by Apache servers, there's no information on the actual availability of the servers—they could have been under a DDoS attack for half that uptime, but as long as they haven't rebooted, they're on the list.

Bottom line: there is no bottom line. For every (seemingly) logical statement, there exists an equally logical retort. We could write the arguments and counter-arguments for twenty pages if we felt like it, and you still won't walk away with a definite conclusion. Let's just move on.

Your OS And You

The biggest security breaches occur when malware is allowed to run with on your system with elevated privileges—which means that it has access to critical programs and data that only your system's kernel should have. Once it's reached that level, your PC becomes its humble servant, and can be brought down at the slightest whim. Who gives this malware its privileges? Well, you do.

With Windows XP, the person who installs the operating system becomes the Administrator, so if you're the only one using your PC, you've got the privileges to wreak all sorts of havoc, should you choose to. Consequently, any application you install and run is also accorded the same royal treatment, no questions asked. Now add to that the fact that Windows' system services run under a user account called SYSTEM (you can check this out in the Task Manager)—the most powerful account on your system, with access to *everything* critical—and that the first processes that malicious programs hijack are system services. You'll be drawing pretty accurate conclusions by now...

Vista, thankfully, changes this. The user who installs Vista is still part of the Administrators group, but even this administrator runs with regular, limited privileges. When administrative tasks—including installing new programs—need performing, User Account Control (UAC) kicks in, telling you that you need to give the task a go-ahead before it, well, goes ahead. If you read the UAC prompt and don't know the program it's warning you about, you can prevent it from running. But what if you've blindly allowed the task to continue?

Unlike XP, malware has little or nothing to gain from hijacking a system service in Vista—instead of giving services unlimited rights like before, Vista runs them with low privileges, giving them access only to critical files and Registry entries that they need; each service is given a Security ID (SID), which it uses to request permission to these objects. Consider this scenario in XP: a virus has hijacked your system's innocuous Plug-and-Play service and has acquired its privileges. One of the first things viruses do is disable the anti-virus so they can

continue their reign of terror uninterrupted. Since the virus can now do anything to your system, it'll be allowed to kill the anti-virus service, corrupt the virus definitions, modify its Registry entries and much more. In Vista, if this same virus manages to attack the service, all damage is restricted to the service and its domain—you won't necessarily be happy, but you won't be feeling silly, either.

Windows Vista basically accomplishes what Linux has been doing for a long time—ensure that regular users don't have access to critical operating system files and services; they don't even have access to files owned by each other! The all-powerful root account is never used, and in Ubuntu, it's actually disabled: you're expected to use the "sudo" command (or gksu for GNOME applications) and enter your password to execute commands that affect the system in general. Unless you *deliberately* use these commands to run a suspicious executable, the worst it can do is affect the files you have access to. Even if, for whatever inexplicable reason, you do decide to run a malicious script with sudo, your services are still safe...

Services in Linux run as separate users, with access only to files that they own; more often than not, they don't even have the rights to use the terminal, so they can't run commands or start other services. This is where the multi-user approach comes handy again—since users are isolated from each other, services can't access the data used by other services. The Apache server, for instance, runs as a user called www-data, which only has access to the Web pages it serves. If a hacker exploits an Apache vulnerability to get into the www-data user account, he can't really do much to the other services, because www-data doesn't own those files. He can, however, mess with Web pages, so while this isn't a doomsday scenario, it's certainly not ideal.

You've probably noticed the parallels between the two operating systems by now—for your use at home, both systems are quite secure. Unfortunately, they both have one critical flaw: PEBKAC—Problem Exists Between Keyboard And Chair. No operating system can plan for human ignorance—Vista, unfortunately, allows you to disable UAC (and it's annoying enough for you to want to), and even then, it can't guard against people who click OK without realising what they're doing. Linux developers do all they can to ensure that you never feel the need to use the root account, but there's always the daredevil who can't resist. If you keep your head on your shoulders and read what's on your screen, your PC will be as secure as you want it to be.

True, there are more people working on Windows viruses than Linux, but popularity isn't all that makes Windows popular with the hacker community...

Package Deal

Unfortunately, the thing we love about Windows is also its undoing. Windows is designed as a single unit, so a lot of features (or modules, depending on what you're looking at) are inte-

grated into the core of the system, even though they don't need to—IE and Outlook Express immediately spring to mind. Remember the Windows 98 + IE4 combination, when you couldn't tell where the OS ended and the browser began?

The upside to this approach is a smooth, hassle-free user experience—all these programs get a little extra love from the OS kernel, so they'll inevitably perform better than third-party programs. The downside—which we see often enough—is that any vulnerability in these programs can potentially bring down the entire system. This is why you hear of vulnerabilities in IE and Office letting hackers gain control over Windows PCs. In fact, in Windows XP, even the graphics engine that draws windows on your screen runs in this privileged space, so if your graphics freeze, your only option is to reset your system.

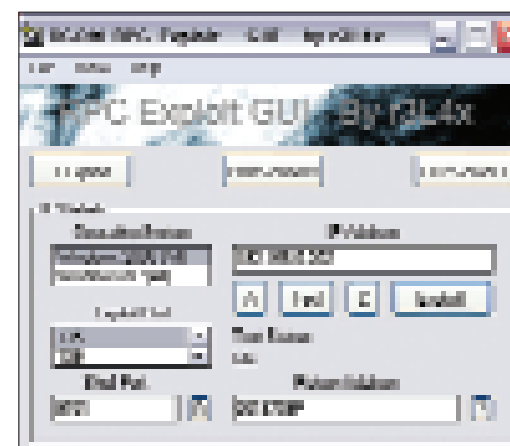
Linux, on the other hand, is designed to be modular. Only the most essential components of the operating system are integrated into its core—the rest are loaded only when needed, and don't have access to critical system files. A vulnerability in Apache, for example, stays in Apache. The Linux we're talking about here is the core OS, and not the Desktop Environments, mind you. GNOME and KDE may not integrate with the Linux kernel, but within themselves, they are quite integrated. The disadvantage is the same as for Windows—if one component of GNOME fails, the Desktop Environment fails, taking all the applications running in it down. However, the OS kernel isn't affected, so all you have to do is a simple [Ctrl] + [Alt] + [Backspace], which ends your GNOME / KDE session and lets you log in to your system again.

This is where Linux (and UNIX, and the BSDs) clearly take the lead in terms of security. It may not feel as nice as Windows while you're using it, but it won't let something silly like a flaw in a text editor take it down, either.

Endgame

So really, which is better? Let's look at it step by step, starting with the potential for malicious software to enter your PC. Both systems are on equal footing here—malicious software getting into your PC is usually your fault (assuming you don't have security software installed). Step two: malicious software does get into your PC; what is the scope of the damage it can do? Again, with both Linux and Vista, damage caused by malware is restricted to the service it exploits, and the files that the service can access. So far, so good. Finally, step three: what happens when the malware goes about its dirty deed? With Vista, if a critical service—like the Remote Procedure Call (RPC) service—is compromised, all manners of chaos may ensue. Every application under Windows needs to use RPC, so you're sunk without it. With Linux, services aren't as tightly integrated with the OS, so while your Linux PC can be crippled—some applications won't run, you may not have network access and so on—the kernel is still safe, which means that with a

Only the most essential components of the operating system are integrated into the Linux core



Windows' Remote Procedure Call (RPC) service has many things working against it—nearly every application on your PC needs it, so it's a favourite target for hackers, and even a moderately resourceful amateur will stumble upon neat little GUI tools like this to exploit the service

little root wizardry, it can be brought back to life again.

Bottom line: for daily desktop use, both systems are equally secure—but if things do go wrong, they go more wrong with Windows. ■

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Technical Specification	
CPU	733 MHz (Upgradable upto Pentium 4 Core 2 Duo)
Memory	128 MB (Upgradable)
Flash Memory	Optional
Embedded OS	Optional (Linux, WinCE, WinXP)
Graphic	15 pin D type female VGA connector
Ethernet Interface	10/100Mbps, RJ45 type cabling
Wireless	Optional
Hard-disk	(Optional) 2.5" Laptop HDD
I/O	1 Parallel, 4 USB, 1 Serial, PS2 Keyboard & Mouse con.
Audio	Ac97 audio
Power	+12V @ 5A (0 watts only)
Dimensions	220" x 9.5" x 9"
Weight	1.5 Kg
Platform supported	K8M, K8A, X86, VNC, NX, Intel, ssh
OS supported	Win NT/2K/XP, Linux, DOS
Local device supported	Parallel/USB & Serial printers, Pen-drives, CD-ROM, Floppy, devices like bar code scanner etc.